

AYURVEDIC DIETARY RECOMMENDATION MODULE

25-26J-305

Project Proposal Report

W.A.N.M. Dias

B.Sc. (Hons) Degree in Information Technology.

Department of Information Technology

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DECLARATION

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I hereby declare that this individual research project titled "Ayurvedic Dietary Recommendation Module" is my own work and that all sources of information have been acknowledged by means of complete references. This research project has been prepared in partial fulfillment of the requirements for the Bachelor of Science (Hons) in Information Technology and has not been submitted for any other degree or diploma at any other university or Institute. I affirm that the data and research findings presented in this document are true to the best of knowledge and belief.

Name	Student ID	Signature
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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

.....me.....

Signature of the Supervisor

...01/09/2025

Date

DEDICATION

This work is dedicated to our supervisor, Ms. Manori Gamage; our co-supervisor, Ms. Uthpala Samarakoon; and to all the friends and family whose guidance, encouragement, and support have been invaluable throughout this journey.

ACKNOWLEDGEMENT

I would like to extend my heartfelt gratitude to our supervisor, Ms. Manori Gamage, our co supervisor, Ms. Uthpala Samarakoon, for their generous time, guidance, and support throughout this journey. Your encouragement and expertise have been invaluable in shaping my work and helping me overcome challenges.

ABSTRACT

Ayurveda, an ancient system of holistic medicine, emphasizes personalized dietary planning based on the individual's constitution (Prakriti), dosha balance, disease condition and contextual features such as season, digestion, and emotional state of the individual. While modern nutritional systems rely on caloric analysis, they lack the depth of qualitative assessment inherited with Ayurvedic principles. This research proposes the development of a context-aware Ayurvedic Dietary Recommendation System that integrates classical Ayurvedic knowledge with computational techniques to deliver personalized meal plans.

The system will extract dosha food relationship and disease specific dietary guidelines from classical Ayurvedic texts and diet manuals, creating a structured food dataset. A rule-based engine will be developed to classify food based on the rasa, virya, vipaka, guna, and pathya-apathya principles of Ayurveda while the adaptive logic will refine recommendations using user feedback on digestion, mood, sleep, and overall satisfaction.

The proposed module aims to bridge traditional Ayurvedic dietary wisdom and modern personalized methods, offering a novel, holistic approach to preventive and therapeutic nutrition.

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1. INTRODUCTION

1.1 Background & Literature survey

Classical Ayurvedic texts such as Charaka Samhita and Ashtanga Hridaya provide elaborate guidelines on food classification, compatibility, seasonal regimens (Ritucharya) and disease-specific dietary prescriptions [1]. These guidelines recognize and dynamic nature of dietary suitability, which varies based on factors such as constitution (Prakriti), digestive strength (Agni), age, season, emotional state, and disease condition. [4][6]

1.1.1 Analysis in Ayurvedic Dietary Systems

Ayurvedic dietary analysis is rooted in a comprehensive understanding of the interaction between food, body constitution (Prakriti), and environmental context. Unlike modern nutrition, which focuses heavily on caloric and macronutrient content, Ayurveda evaluates food based on its qualitative attributes such as rasa (taste), guna (quality), virya (potency), vipaka (post-digestive effect), and karma (therapeutic action) [4] [5]. These multi-dimensional parameters are analyzed to determine a food's compatibility with an individual's dosha and its capacity to either pacify or aggravate imbalances.

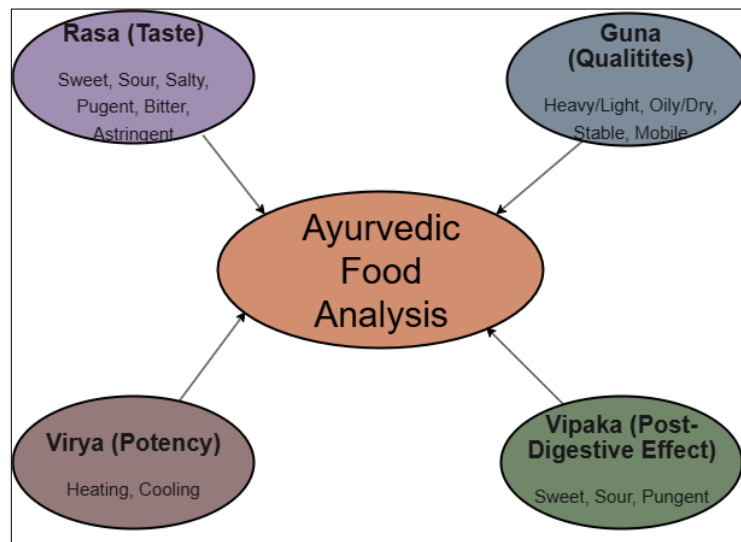


Figure 1-Ayurvedic Food Analysis

Furthermore, disease-specific and seasonal analysis plays a vital role in Ayurvedic dietary planning. For instance, bitter and astringent foods are recommended in spring to pacify Kapha, while sweet and cooling foods are favored in summer to soothe aggravated Pitta

[4]. This detailed contextual analysis provides a holistic framework where the same food may be considered pathya (wholesome) or apathya (unwholesome) based on factors like the user's digestive strength (Agni), emotional state, current illness, age, and climate [1].

Modern implementations of Ayurveda-inspired diet systems often simplify this complexity, typically scoring ingredients based on binary suitability for dosha types [3]. However, a full Ayurvedic analysis integrates these variables dynamically, allowing food recommendations to shift contextually.

1.1.2 Dosha–Food Relationships and Classical Principles

Svasthya, “to be established in oneself or own natural state” is optimal health according to Ayurveda. In order to achieve this, one must have a balance of structural and physiological factors, metabolic and excretory processes, body tissues, senses, mind, and attain a state of self-awareness and contented self [1][5] Ten factors (dasa vidha pariksa) are used to determine the state of health of an individual per Ayurveda namely, body tissues (dusya), residing location (desa), physical strength (bala), seasons/time (kala), digestive and metabolic processes (agni or anala), genetic and phenetic constitution (prakriti), age (vaya), mental strength or temperament (sattva), habituation (satmya), and food (ahara)[2]. It is interesting to note that sattva is said to influence health. While these are mainly used for diagnostic purposes, these can also be used to measure the wellbeing of an individual.

1.1.3 Machine Learning in Ayurvedic Dietary Personalization

The integration of machine learning (ML) techniques into Ayurvedic dietary systems represents a growing trend to modernize and scale personalized health recommendations. While traditional Ayurvedic recommendations are inherently personalized based on a user's dosha type, digestive strength, and contextual factors, ML models aim to replicate and enhance this personalization through data-driven algorithms.

Recent works, such as the Ayurveda-Based Food Recommendation System by Ismayilova et al. (2024), demonstrate how ML can be used to bridge classical Ayurvedic principles with modern recommendation engines [3]. This system employed Natural Language Processing (NLP) to extract ingredients from recipes and used a K Nearest Neighbors (KNN) algorithm to recommend dishes aligned with the user's dosha type. The ML model calculated a dosha compatibility score based on whether ingredients were listed as "favor" or "avoid" for each dosha, producing personalized recipe lists [3]. However, challenges

such as dataset imbalance and limited contextual adaptability (e.g. not accounting for digestion or mood) were noted.

In contrast, the proposed system introduces a hybrid approach, combining rule-based Ayurvedic logic with ML driven adaptation mechanisms. While the initial classification and filtering of foods will rely on verified dosha food mappings, ML can be leveraged to learn user preferences and dynamic attributes over time. For example, models could be trained in user feedback (e.g. ratings on digestion, mood, or satisfaction) to identify subtle patterns and optimize weekly diet plans. This creates a closed-loop recommendation system where ML supports continuous personalization beyond static dosha profiles [6].

Aspect	Traditional Ayurvedic Literature	Existing Research Systems	Proposed System
Knowledge Source	Charaka Samhita, Ashtanga Hridaya, manuals	Limited digitized datasets	Classical texts + validated manuals + mined recipes
Diet Classification	Based on rasa, virya, vipaka, guna, pathya/apathya	Mostly binary dosha-food mapping	Multi-attribute rule-based classification
Disease Integration	Pathya/apathya for specific diseases	Often ignored	Explicit disease-wise dietary recommendations
Personalization	By practitioner (Prakriti, Agni, season)	Basic dosha-type personalization	Context-aware: digestion, mood, sleep, age, season
Machine Learning Use	Not applicable	NLP + KNN for recipe recommendation [3]	Hybrid ML (KNN, RF, SVM) + adaptive feedback
Validation	Practitioner expertise	Minimal expert involvement	Practitioner-validated dataset + user feedback

Table 1-ML Overview

1.1.4 Data Mining in Ayurvedic Dietary Recommendation Systems

Data mining plays a crucial role in transforming the qualitative, rule-rich domain of Ayurveda into actionable, structured datasets suitable for computational modeling. In the context of Ayurvedic dietary systems, data mining involves extracting, organizing, and categorizing large volumes of dietary information both from classical texts and modern recipe datasets to enable meaningful analysis and personalized recommendations [3][6].

Recent implementations have demonstrated how data mining techniques can be applied to large recipe datasets to derive Ayurvedic relevance. For instance, Ismayilova et al. (2024) mined over 350,000 recipes from Food.com and AllRecipes.com, then filtered and refined them to retain 28,000 nutritionally complete records. These were annotated with Ayurvedic dosha mappings using NLP-based ingredient extraction and lookup tables [3]. The authors created a scoring system by matching ingredients with dosha-specific lists to evaluate recipe compatibility, thus enabling downstream ML-driven recommendations.

The application of association rule mining and pattern discovery can uncover co-occurrence of symptoms, foods, and seasonal or lifestyle factors in Ayurvedic datasets. These insights can inform rule-based engines and enrich ML models with deeper contextual understanding [6]. For example, identifying common diet patterns in Kapha dominant individuals during winter may support both preventive and therapeutic recommendations.

1.2 Research Gap

Despite the advancement of integrating Ayurvedic principles into digital dietary systems, several limitations remain in existing research.

Current systems primarily focus on general health guidelines without adapting meal plans to real time user conditions such as digestion, mood and sleep patterns. [6] This module introduces context-aware personalization, dynamically adjusting based on the human health parameters evolving their needs.

While some studies Ayurvedic concepts, few implement a comprehensive dosha centric model that considers rasa, virya, guna and Prabhava collectively to ensure holistic meal compatibility [5] and there is an inadequate use adaptive learning where most ayurvedic dietary systems lack machine learning mechanisms that evolve through user feedback. [6]

Few existing ayurvedic dietary recommendation systems incorporate direct validation from certified ayurvedic practitioners during the development and testing phase. Most solely rely on textual interpretations of classical ayurvedic literature or datasets without

expert review, leading to potential inaccuracies in food dosha disease relationships.[4] The absence of this crucial step In prior to development highlights the need for models which integrates the practitioner expertise at stages of dataset annotation and verification.

1.3 Research Problem

Existing systems exhibit several limitations that hinder their effectiveness in delivering holistic and personalized nutrition guidance. The primary research problems identified are as follows,

1. Limited personalization in Existing Models

Most available dietary recommendation systems offer only static meal suggestions, typically based on broad dosha categorizations. These models fail to consider dynamic user attributes of humans such as digestion quality, mood, sleep patterns, age, and changing health conditions [3][6]. Without lack of adaptation, recommendations lack accuracy and fail to replicate the individualized care provided by Ayurvedic practitioners.

2. Incomplete Dosha based Integration

While Ayurvedic concepts like Vata, Pitta, and Kapha are referenced in some studies, their application relies only on simple favor/ avoid food lists. [7] True Ayurvedic dietary planning involves a deeper evaluation rasa (taste), guna (qualities), virya (potency), vipaka (post-digestive effect), and disease-specific contradictions. The absence of this complete frame of parameters results in incomplete representation of Ayurvedic principles in existing computational frameworks.

3. Insufficient Disease Specific Dietary Mapping

Ayurvedic dietary guidance extends beyond general wellness and emphasizes disease-specific dietary protocols (*pathya-apathya*) for managing conditions such as diabetes, rheumatoid arthritis, and acidity [8]. However, current models rarely incorporate disease-oriented dietary frameworks validated against classical texts. This leads to generic meal suggestions, limiting the therapeutic value of these systems.

4. Lack of Practitioner- Validated Computational Models

Few studies involve Ayurvedic practitioners in reviewing datasets, verifying rule-based engines, or validating model outputs [8]. This omission risks inaccuracies,

misinterpretation of Ayurvedic principles, and reduced clinical reliability. Practitioner input is essential to ensure authenticity, contextual accuracy, and the safe application of Ayurvedic dietary logic in real-world scenarios.

Central Research Problem:

The lack of comprehensive personalization, providing static meal suggestions and focusing narrowly on dosha based food balances without practitioner validation results in incomplete representation of Ayurvedic principles and limited clinical relevance.

2. OBJECTIVES

2.1 Main Objective

The primary objective of this component is to develop a personalized Ayurvedic Dietary Recommendation System that aligns classical Ayurvedic food principles with modern rule-based and data-driven techniques that integrate classic Ayurvedic knowledge with adaptive computational techniques to deliver precise, dynamic, practitioner validated meal plans.

2.2 Specific Objectives:

- Extract and structure food-dosha relationships from classical Ayurvedic texts and validated dietary manuals through systematic review, annotation, and organization of dietary principles, ensuring accurate alignment with Ayurvedic concepts such as Prakriti, Agni, and seasonal variations.
- Build a food classification rule base incorporating rasa, virya, vipaka, and dosha compatibility, enabling computational representation of Ayurvedic dietary logic for personalized meal generation.
- Design a rule-based engine to generate diet plans based on dosha, disease, and seasonal inputs, integrating classical guidelines with contextual adjustments for preventive and therapeutic nutrition.
- Integrate adaptive logic to modify recommendations based on digestion, mood, sleep, and age, leveraging dynamic user feedback to personalize meal plans in real-time.
- Implement user feedback mechanisms to support ongoing personalization, capturing user ratings on meal suitability, digestion quality, and overall satisfaction for iterative system refinement.
- Consult Ayurvedic dieticians to validate food mappings and refine contextual accuracy, ensuring that computational outputs are clinically relevant and aligned with authentic Ayurvedic practice.
- Evaluate system effectiveness through accuracy, usability, and user satisfaction metrics, employing comparative testing against baseline models to demonstrate improvement in personalization and adherence to Ayurvedic principles.

3. METHODOLOGY

3.1 System Architecture Overview

The Ayurvedic Dietary Recommendation System is designed to integrate classical Ayurvedic dietary principles with modern computational methods, delivering personalized, context-aware meal plans. The system architecture comprises three core layers: Data Layer, Processing Layer, and Output Layer, ensuring structured data handling, intelligent analysis, and adaptive recommendation generation.

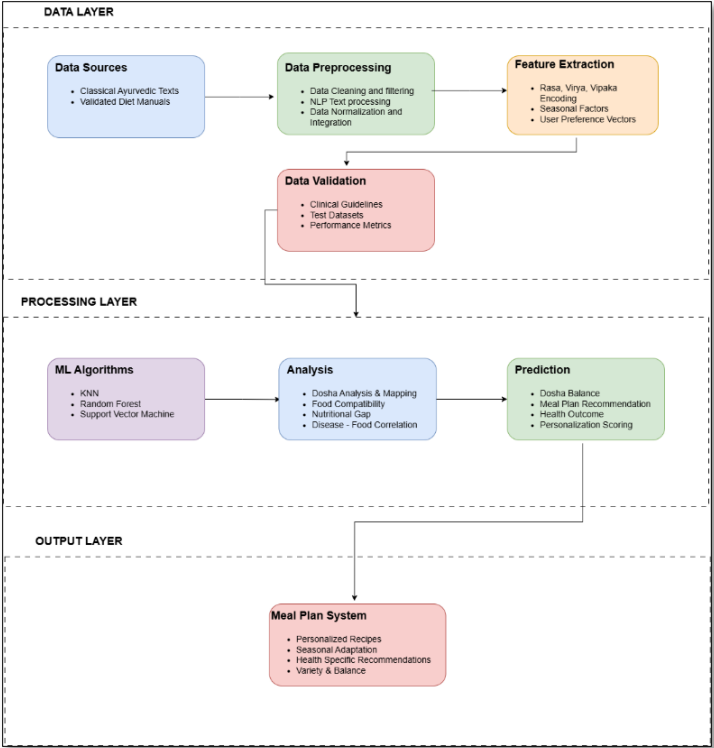


Figure 2-Ayurvedic Dietary Recommendation System Overview

3.1.1 Data Layer

This layer forms the foundation of the system, collecting raw data and preprocess information essential for accurate recommendations.

Data sources

The system derives information from classical ayurvedic texts, such as Charaka Samhita and Ashtanga Hridaya, along with validated ayurvedic diet manuals. These sources provide insights into food classification, dosha compatibility, rasa (taste), virya (potency), vipaka (post-digestive effect), and disease-specific dietary guidelines (pathya-apathya).

Data Preprocessing

The extracted data undergoes cleaning and filtering to remove noise and inconsistencies, followed by Natural Processing Language techniques to interpret and digitize textual knowledge. Data normalization ensures compatibility with computational models, and integration combines multiple sources into a unified dataset.

Feature Extraction

Key Ayurvedic attributes such as rasa, virya, vipaka, and guna are encoded as features, alongside contextual variables like seasonal factors and user preference vectors. These features enable a nuanced representation of dietary recommendations beyond simple favor/avoid lists.

Data Validation

Validation mechanisms, including expert review and testing datasets, ensure the reliability of the mappings while performance metrics are used to evaluate consistency across datasets to maintain clinical relevance.

3.1.2 Processing Layer

The Processing Layer is responsible for intelligent analysis, rule application, and adaptive personalization of dietary recommendations.

3.1.2.1 Machine Learning Algorithms

Algorithms such as K-Nearest Neighbors (KNN), Random Forest, and Support Vector Machines (SVM) are applied to enhance recommendation accuracy. These models learn patterns from user feedback and historical data to refine predictions over time.

K-Nearest Neighbors (KNN)

KNN is particularly effective for similar-based recommendations.

When a user provides their profile (dosha constitution, digestion quality, age, seasonal context, etc.), KNN compares it with existing user profiles and identifies the "nearest neighbors" with similar attributes.

Random Forest

Random Forest is a robust ensemble method that handles complex decision-making.

It can be used to classify foods based on multi-dimensional Ayurvedic attributes (rasa, virya, vipaka, guna) and contextual factors (season, disease condition, mood, sleep). Multiple decision trees evaluate whether a food is “suitable,” “neutral,” or “to be avoided” for a user’s dosha-disease profile.

Support Vector Machines (SVM)

SVM is valuable for pattern recognition and classification in high-dimensional feature spaces.

It can be applied to detect subtle patterns in user feedback and Ayurvedic parameters. SVM can classify meal plans into categories such as “highly suitable,” “moderately suitable,” or “unsuitable” based on feedback on digestion, mood, and health outcomes.

Algorithm	Expected Accuracy	Training Time	Constitutional Feature Handling	Advantages
K-Nearest Neighbors (KNN)	Moderate (70–80%)	Low (fast training, lazy learner)	Can map user similarity based on dosha profile, digestion, mood, and seasonal inputs	Simple to implement, effective for similarity-based recommendations
Random Forest	High (80–90%)	Moderate (parallel training possible)	Handles multiple Ayurvedic features (rasa, virya, vipaka, guna) and disease-food mappings effectively	Robust to overfitting, works well with multi-dimensional data
Support Vector Machine (SVM)	High (85–90%) for well-structured features	High (longer training time for large datasets)	Effective for identifying subtle patterns in feedback-based features (e.g., digestion quality, mood scoring)	Good for high-dimensional data, strong generalization
Hybrid/Ensemble (KNN + RF + SVM)	Very High (90%+)	High (depends on integration)	Integrates both rule-based and feedback-based features for holistic personalization	Combines strengths of all models, improved accuracy and adaptability

Table 2-Algorithms & Features

Analysis Module

This performs multi-dimensional evaluation of food compatibility, dosha balance, nutritional adequacy, and disease specific correlations. It ensures that the recommendations follow the ayurvedic guidelines while addressing the individual needs.

Prediction Module

Based on processed data and analytical outcomes, the prediction component generates personalized dietary suggestions. These include meal plans tailored to dosha balance, seasonal changes, health outcomes, and user-specific feedback.

3.1.3 Output Layer

This layer translates analytical results into practical, user-friendly recommendations

Meal Plan System

The final output provides daily, and weekly meal plans with personalized recipes, adaptations and disease specific guidelines for the user. Recommendations are designed to maintain nutritional balance, variety and adherence to ayurvedic principles.

User Centric Feedback Integration

The module incorporates user feedback such as digestion quality, mood, and satisfaction scores to refine future meal plans, creating a closed-loop recommendation system.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

1. User Profile Management

The system must allow the user to register and input their personal details such as age, gender, lifestyle habits along with the medical history.

The system must capture the constitutional features through a questionnaire or practitioner input.

2. Data Input and Preprocessing

The system shall extract and preprocess dietary knowledge from classical Ayurvedic texts and validated manuals.

The system shall normalize food data using Ayurvedic attributes (rasa, virya, vipaka, guna) and contextual features such as season, mood, digestion and sleep.

3. Food Classification and Mapping

The system must classify food into different categories based on dosha compatibility and disease specific guidelines and consistently maintain a structured food disease dosha dataset for rule-based decision making.

4. Recommendation Engine

The system must generate personalized meal plans daily and weekly based on the user parameters, conditions and seasonal context by incorporating machine learning models for adaptive personalization.

The system should provide diet modifications for specific health conditions and integrate with practitioner reviewed diet mappings.

5. Data Validation

The system must enable ayurvedic practitioners to review and validate food dosha-disease mappings and ensure accuracy through dataset cross validation and expert consultation.

4.2 Non-Functional Requirements

1. Performance

The system shall generate dietary recommendations within 3–5 seconds.

2. Scalability

The system shall support increasing users and datasets without performance loss.

3. Usability

The interface should be simple, intuitive, and accessible to non-technical users.

4. Reliability

The system shall ensure consistent recommendations with $\geq 95\%$ uptime.

5. Accuracy

Recommendations shall align $\geq 85\%$ with practitioner-reviewed guidelines.

6. Security

User health data shall be stored securely with encryption and access control.

7. Maintainability

The system should allow easy updates to datasets, rules, and models.

5. PROJECT TIMELINE

The research project will be implemented within a 12-month period to ensure systematic progress allowing iterative refinements and validation. The project will be implemented according to the Gantt chart below which illustrates the distinct project phases with timelines. This timeline ensures systematic progress while allowing iterative refinement through overlapping phases of development, validation, and evaluation.

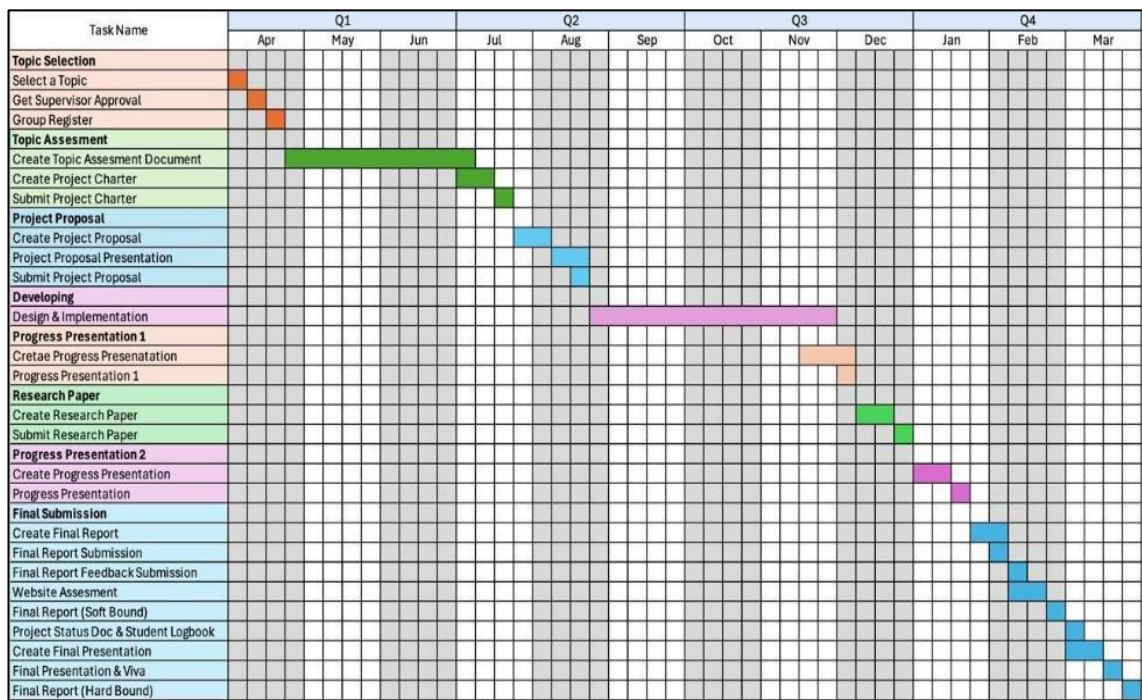


Figure 3- Project Timeline Gantt Chart

6. DESCRIPTION OF PERSONAL AND FACILITIES

My academic background in Information Technology provides a solid foundation for undertaking this research project. With a strong interest in machine learning, artificial intelligence and human computer interaction, I have developed skills in building intelligent systems that merge traditional knowledge with modern computational approaches. I am proficient in Python programming and well-knowledgeable in machine learning libraries such as Pandas, TensorFlow which are essential for developing adaptive algorithms and performing large scale data analyzing. Module coursework and projects in data mining, natural language processing, and recommendation systems have strengthened my ability to extract, preprocess and classify datasets, in this case, Ayurvedic food-dosha mappings.

Domain-specific exposure is brought through independent study of Ayurvedic dietary principles and holistic health practices which enables me to interpret classical texts in alignment with computational logic for the system. This contextual knowledge will be collaborated with Ayurvedic practitioners and ensure that the system remains authentic and relevant. A hybrid recommendation framework with rule-based systems and machine learning prototypes with adaptive personalization will be developed with the existing and add on knowledge.

The research will be supported by the technical and academic facilities at SLIIT, which include access to digital libraires, high-performance computing laboratories with modern hardware and software environments. These resources enable model training, preprocessing and real time system testing. Development will be carried out using industry standard tools such as Python IDEs, Git for version controlling and cloud-based environments for deployment and testing. These facilities ensure reliability and efficiency for handling ayurvedic datasets and machine learning integration.

Together with technical expertise and background knowledge and access to advanced facilities, provide the necessary capabilities to successfully design, implement and validate the Ayurvedic Dietary Recommendation Module.

7. BUDGET AND BUDGET ALLOCATION

Data Collection and Processing: LKR 50,000

- Travel expenses for hospital visits and expert consultations
- Digitization and annotation costs for classical texts
- Survey distribution and data collection materials

Software and Computing Resources: LKR 25,000

- Cloud computing resources for model training
- Specialized software licenses for text processing
- Database hosting and management costs

Expert Consultation Fees: LKR 75,000

- Ayurvedic practitioner consultation fees
- Sanskrit expert annotation services
- Validation study compensation

Research Materials and Documentation: LKR 15,000

- Classical text purchases and access fees
- Reference materials and research publications
- Documentation and report preparation

Total Budget: LKR 165,000

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